

Project Report – Best Weather App in the World Mate

1. Title & Course Information

Project Title: Best Weather App in the World Mate

Course: Python Flipped Course (VITyarthi Platform)

Component: Evaluated Course Project

2. Objective

The project aims to build a Python-based weather application that fetches real-time weather data using an API and displays it in a simple, user-friendly format via the command line.

3. Features Implemented

Fetches live weather data from API (OpenWeatherMap or similar).

Command-line interface for city input.

Error handling for invalid city names or API failures.

Displays temperature, humidity, and weather conditions.

Secure API key management using .env file.

4. Setup & Execution

Clone the repository:

```
bash
```

```
git clone https://github.com/vyanktreshgour-sudo/best-wather-app-in-the-world-mate
```

```
cd best-wather-app-in-the-world-mate
```

Install dependencies:

```
bash
```

```
pip install -r requirements.txt
```

Create a .env file in the project root with:

Code

```
WEATHER_API_KEY=2fd17f1979692b6744a34e9ed52fa2d5
```

Run the app:

```
bash
```

```
python main.py --city "London"
```

5. Screenshots & Demo

Screenshots: Available in screenshots/ folder.

Screen Recording: Linked in README.md.

6. Challenges Faced

Handling API errors and invalid inputs.

Ensuring smooth command-line execution.

Managing environment variables securely.

7. Future Improvements

Add support for multiple cities at once.

Integrate forecast data (next 5 days).

Improve CLI output formatting with colors.

Add unit tests for reliability.